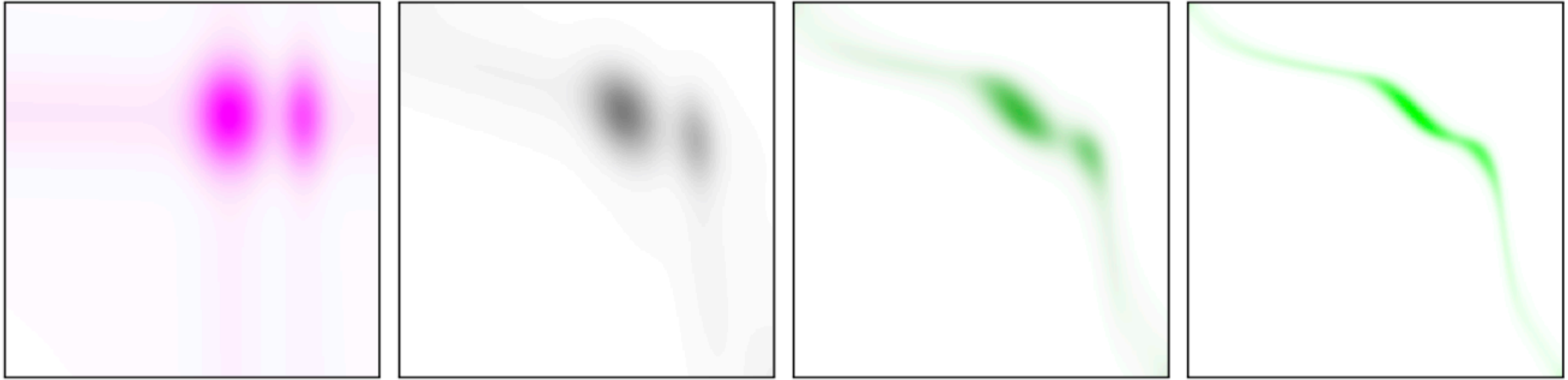
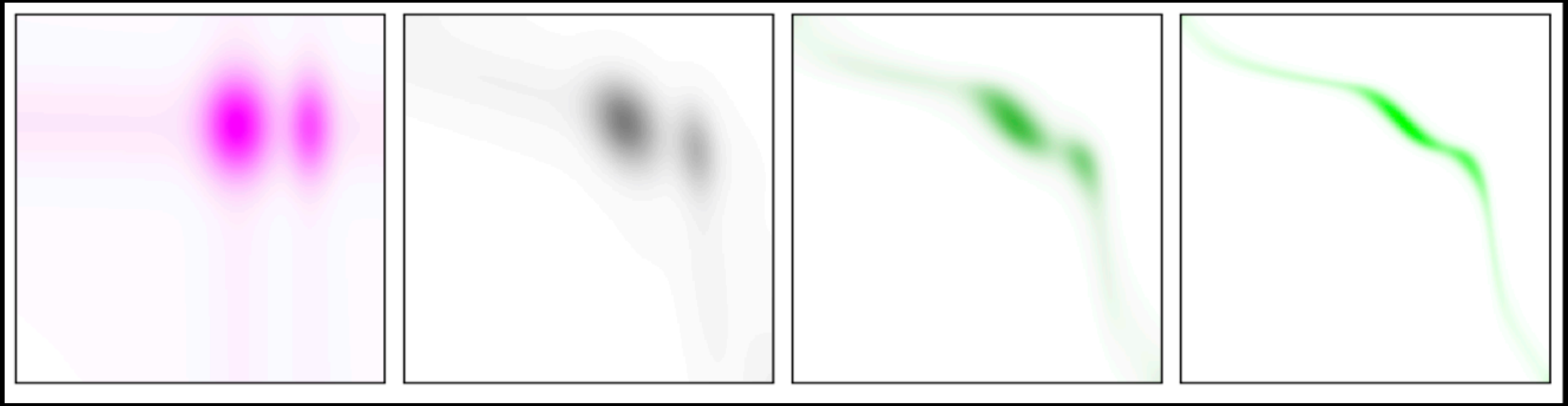
Quantifies how dense/sparse



Entropy of π

- We modify the primal problem by adding a regularization term:
  $H(\pi) = -\sum_{ij} \pi_{ij} \log \pi_{ij}$ (= is the entropy of π)
- Alternatively, can use relative entropy: $\mathrm{KL}(\pi \mid \mathbf{a} \otimes \mathbf{b}) = -\sum_{ij} \pi_{ij} \log \frac{\pi_{ij}}{a_i b_j}$

High $H(\pi)$: Low $H(\pi)$:



- **Quantifies how dense/sparse** π is (= how much *mass splitting* it implies)

Entropic OT Formulation